

## FK SERIES ■ HIGH VOLTAGE, 105°C TYPE

## KEY FEATURES

- ALUMINUM ELECTROLYTIC CAPACITOR ■ THT type
- High voltage, high reliability



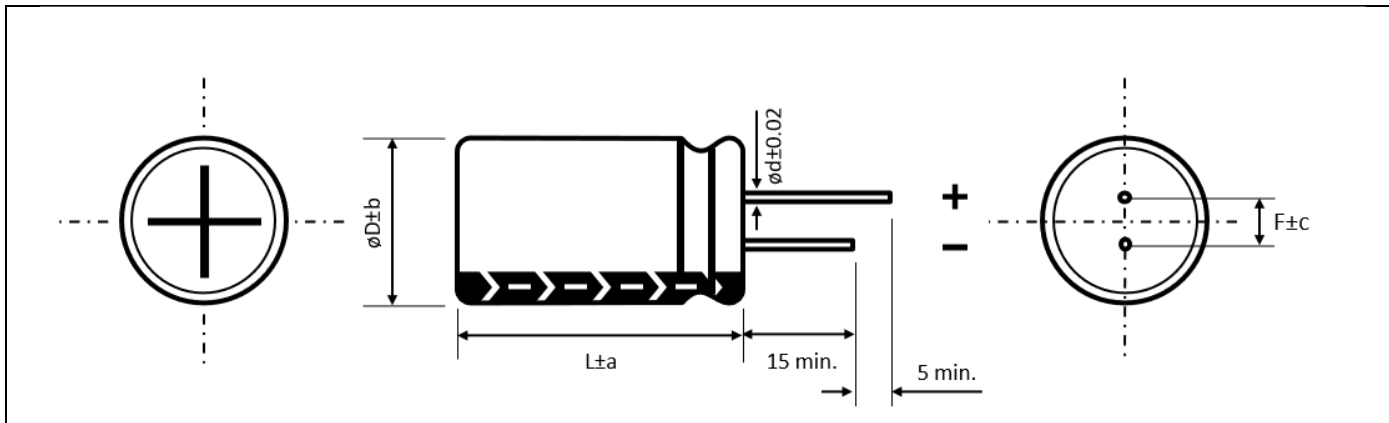
## SPECIFICATIONS

| Part Number                                                                                           |                   | FK100M160G160ETA |     | Unit | Conditions                  |
|-------------------------------------------------------------------------------------------------------|-------------------|------------------|-----|------|-----------------------------|
| Capacitance                                                                                           | C <sub>R</sub>    | 10               |     | μF   |                             |
| Rated Voltage                                                                                         | V <sub>R</sub>    | 160              |     | V DC |                             |
| Surge Voltage                                                                                         | V <sub>S</sub>    | 184              |     | V DC |                             |
| Cap. Tolerance                                                                                        | ΔC                | -20~20           |     | %    | 120Hz ▪ 20°C                |
| Ripple Current                                                                                        | I <sub>R</sub>    | 147              |     | mA   | 120Hz ▪ 105°C               |
| Ripple Current                                                                                        | I <sub>R</sub>    | 368              |     | mA   | 100KHz ▪ 105°C              |
| Dissipation Factor                                                                                    | tanδ              | 15               |     | %    | 120Hz ▪ 20°C                |
| Leakage Current                                                                                       | I <sub>LEAK</sub> | 164              |     | μA   | 1 min ▪ 20°C                |
| Size                                                                                                  | D x L             | 10X16            |     | mm   |                             |
| Pitch                                                                                                 | F                 | 5                |     | mm   |                             |
| Operating Temperature Range                                                                           |                   | -40~105          |     | °C   |                             |
| AEC-Q200 Qualified                                                                                    |                   | NO               |     |      |                             |
| Lifetime Test                                                                                         |                   |                  |     |      |                             |
| Endurance<br>V <sub>R</sub> & I <sub>R</sub> applied                                                  | Test              | 8000             |     | h    | 105°C                       |
|                                                                                                       | ΔC/C <sub>R</sub> | ≤                | ±20 | %    | of initial measured value   |
|                                                                                                       | tanδ              | ≤                | 200 | %    | of initial specified value  |
|                                                                                                       | I <sub>LEAK</sub> | ≤                |     |      | the initial specified value |
| Shelf Life<br>V <sub>R</sub> = 0                                                                      | Test              | 1000             |     | h    | 105°C                       |
|                                                                                                       | ΔC/C <sub>R</sub> | ≤                | ±20 | %    | of initial measured value   |
|                                                                                                       | tanδ              | ≤                | 200 | %    | of initial specified value  |
|                                                                                                       | I <sub>LEAK</sub> | ≤                |     |      | the initial specified value |
| Before measurement: Restore capacitor to 20°C, apply V <sub>R</sub> for 30 min according JIS-C-5101-4 |                   |                  |     |      |                             |

## PART NUMBER SYSTEM

| Series | $C_R$<br>( $\mu\text{F}$ ) | $\Delta C$<br>(%) | $V_R$<br>(V) | D<br>(mm) | L<br>(mm) | Type                | Lead Treatment | Special     |
|--------|----------------------------|-------------------|--------------|-----------|-----------|---------------------|----------------|-------------|
| 1_2    | 3_4_5                      | 6                 | 7_8_9        | 10        | 11_12_13  | 14                  | 15_16          | 17_18_19_20 |
| FK     | 100                        | M                 | 160          | G         | 160       | E                   | TA             |             |
|        | 10 $\mu\text{F}$           | -20~20%           | 160V         | 10mm      | 16mm      | With lead treatment | Tape / Ammo    |             |
|        |                            |                   |              |           |           |                     |                |             |
|        |                            |                   |              |           |           |                     |                |             |
|        |                            |                   |              |           |           |                     |                |             |

## DIMENSIONS ■ In mm. For TAPED VERSIONS, please see our packaging information in the table below.



| $\phi D$ | $b$       | $L$ | $a$       | $F$ | $c$ | $\phi d$ |
|----------|-----------|-----|-----------|-----|-----|----------|
| 10       | $\pm 0.5$ | 16  | $\pm 1.5$ | 5   | 0.5 | 0.6      |

## MULTIPLIER $K_f$ for RIPPLE CURRENT vs. FREQUENCY

| Frequency (Hz) | 60 | 120 | 300 | 400 | 500 | 1k   | 3k   | 10k  | 50k  | 100k | 300k |
|----------------|----|-----|-----|-----|-----|------|------|------|------|------|------|
| $K_f$          |    | 1   | 1   | 1   | 1   | 1.75 | 1.75 | 2.25 | 2.25 | 2.5  | 2.5  |

## PRECAUTIONS, GUIDELINES AND PACKAGING INFORMATION

Unless otherwise agreed in individual specifications, all products are subject to our “General Precautions and Guidelines” as well as our “Packaging Information”. Please refer to the following links in the table.

|                                                      |                                       |                           |                                   |
|------------------------------------------------------|---------------------------------------|---------------------------|-----------------------------------|
|                                                      |                                       |                           |                                   |
| <a href="#">General Precautions &amp; Guidelines</a> | <a href="#">Packaging Information</a> | <a href="#">3D Models</a> | <a href="#">Reliability Tests</a> |

### DISCLAIMER

All product related data (e.g. specification, statements and general information) are subject to change without any notice. It is necessary that the customer observes all product related technical / application information and handling instructions.

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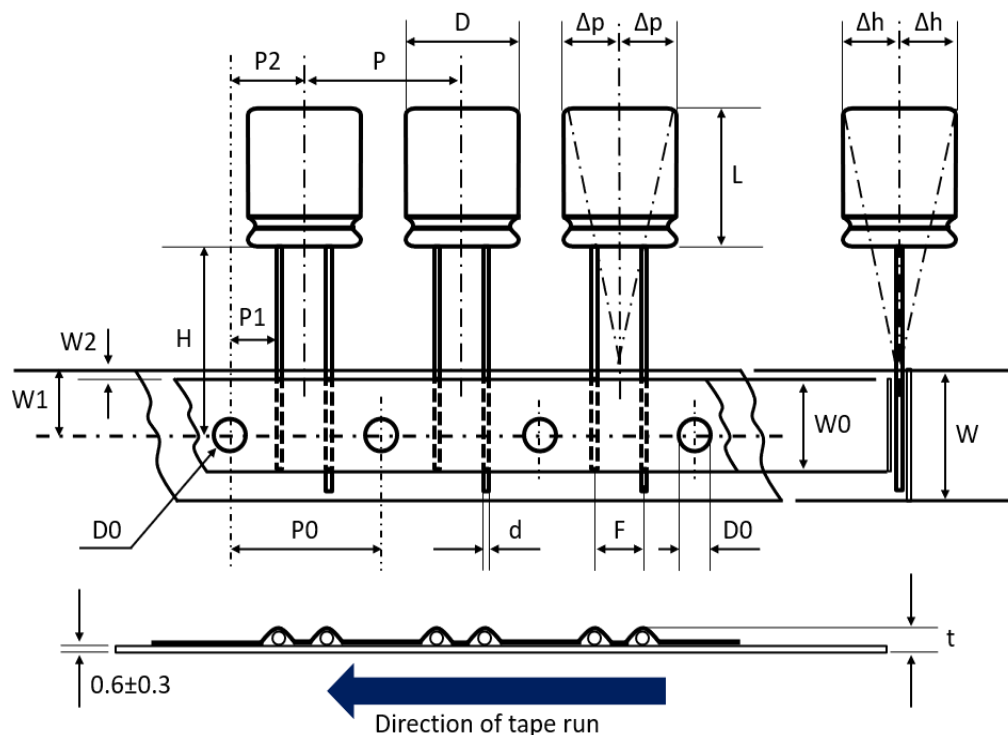
Particular operating conditions (ambient temperature, ripple current, voltage, thermal resistance, etc.) as well as storage, production or assembly may affect the performance and the lifetime of the capacitor. Please consult CapXon for lifetime estimation, failure mode considerations or worst-case scenarios according to the product technology, product tolerances / deviations or change of the characteristics of the capacitor due to shipment, storage, handling, production and usage.

For aerospace or military application, life-saving, life-sustaining, safety critical applications or any application where failure may cause severe personal injury or death, please consult us before design-in the capacitor in your application.

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For further information, please visit our website [www.capxongroup.com](http://www.capxongroup.com) or contact CapXon directly.

## ADDITIONAL SPECIFICATION



|      | P    | P0   | P1   | P2   | F            | W    | W0   | W1   | W2  | H             | D0   | Δh  | Δp  | t   |
|------|------|------|------|------|--------------|------|------|------|-----|---------------|------|-----|-----|-----|
| Tol  | ±1.0 | ±0.2 | ±0.7 | ±1.3 | +0.4<br>-0.2 | ±0.5 | ±0.5 | ±0.5 | Max | +0.75<br>-0.5 | ±0.2 | Max | Max | Max |
| Item | 12.7 | 12.7 | 3.85 | 6.35 | 5            | 18   | 11   | 9    | 2   | 18.5          | 4    | 1   | 1   | 1.5 |

## REVISION RECORDS

| Ver. | Date       | Process Number   | Changed by | Description           |
|------|------------|------------------|------------|-----------------------|
| 001  | 2024-11-02 | TBAKA-2411000002 | Xu XiaoJun | Initial specification |
| 002  |            |                  |            |                       |
| 003  |            |                  |            |                       |
| 004  |            |                  |            |                       |
| 005  |            |                  |            |                       |
| 006  |            |                  |            |                       |

## CUSTOMER APPROVAL

| Date | Name | Department | Signature |
|------|------|------------|-----------|
|      |      |            |           |
|      |      |            |           |
|      |      |            |           |
|      |      |            |           |